ACTS generic detector simulation, no pileup
HNL val (2999 events); previews: single seed, screening budget
fake rate ≤ 0.0294 (pooled HNL val)

ttbar prompt
W muons

L3 triplets, HNL-trained (full budget, v1), MWS-layer + merging (1 seed)
Canto-H (preview): hand-cut helix union, MWS-layer + merging (1 seed)
Canto-L (preview): learned triplet union, MWS-layer + merging (1 seed)
Canto-B (preview): base graph, MWS-layer + merging (1 seed)
ladder: L3 triplets (absolute coords), ttbar, MWS-layer + merging (1 seed)
acorn IGNN, HNL-trained + Canto builder (MWS-layer + merging) (3 seeds)
ladder: L0 (exactly invariant), ttbar, MWS-layer + merging (1 seed)
ladder: blind triplets (no position), ttbar, MWS-layer + merging (1 seed)
ladder: L1 (symmetric + regions), ttbar, MWS-layer + merging (1 seed)
ladder: L0 + trunc., ttbar, MWS-layer + merging (1 seed)
ladder: rz triplets, ttbar, MWS-layer + merging (1 seed)
ladder: L3 + trunc., ttbar, MWS-layer + merging (1 seed)
Canto-B (full budget): base graph, MWS-layer + merging (2 seeds)
acorn IGNN, ttbar+HNL-trained + Canto builder (MWS-layer + merging) (3 seeds)
acorn IGNN, ttbar-trained + Canto builder (MWS-layer + merging) (3 seeds)
ladder: IGNN (acorn arch.), screening budget, ttbar, MWS-layer + merging (1 seed)
ladder: IGNN + trunc., ttbar, MWS-layer + merging (1 seed)
acorn IGNN, ttbar+HNL-trained + CC (prereg. cut rule) (3 seeds)
acorn IGNN, HNL-trained + CC (prereg. cut rule) (3 seeds)
acorn IGNN, ttbar-trained + CC (prereg. cut rule) (3 seeds)
DeZoort IN-L (127k par.) + CC [FR* unreachable] (1 seed)
DeZoort IN (6.4k par.) + CC [FR* unreachable] (1 seed)

0.60 0.65 0.70 0.75 0.80 0.85 0.90 0.95 1.00

Primary-population track efficiency (bars), DM